

ZULFIQAR KHAN

Data Scientist | Deep Learning, NLP & Quantitative Finance

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PROFESSIONAL SUMMARY

AI and Data Science professional with 5+ years of experience deploying machine learning solutions across healthcare, education, and finance. Fluent in three languages, with a track record of community leadership and cross-functional collaboration. Currently completing a BSc Data Science and AI (Top-Up) at the University of the West of England, Bristol.

SKILLS

Programming & Data Analysis: Python, R, SQL, Pandas, NumPy, Scikit-learn, SciPy, Statsmodels, PySpark, PostgreSQL, MongoDB

Machine Learning & AI: TensorFlow, PyTorch, NLP, Computer Vision, Reinforcement Learning, Feature Engineering, Hyperparameter Tuning, Model Evaluation, Deep Learning, Neural Networks, Transformers, LLMs, Attention Mechanisms, Ensemble Methods, Anomaly Detection, Transfer Learning

Data Visualization & BI: Matplotlib, Seaborn, Plotly, Tableau, Power BI

Deployment & DevOps: Flask, FastAPI, Docker, Git, REST API, Linux, YAML, CI/CD

Finance & Quantitative Methods: Portfolio Optimisation, Time Series Analysis, Risk Modelling, Walk-Forward Validation, Factor Modelling, Sharpe Ratio Analysis, Monte Carlo Simulation, Derivatives Pricing, Asset Allocation Strategies

Professional Skills: Research Leadership, Cross-functional Communication, Stakeholder Presentation, Mentorship, Event Organisation, Critical Analysis, Academic Writing, Public Speaking

PROFESSIONAL EXPERIENCE

Machine Learning Researcher | IMSciences, Peshawar *October 2020 – May 2025*

- Developed a deep learning diagnostic model for early disease detection achieving 93% accuracy, cutting manual review time by 40% across 50,000+ patient records.
- Engineered end-to-end machine learning pipelines for an educational integrity system, flagging anomalies across 10,000+ student submissions at 89% precision.
- Accelerated model deployment cycle time by 35% through containerisation with Docker and FastAPI.
- Conducted feature engineering, hyperparameter tuning, and model evaluation across supervised and unsupervised learning tasks.
- Mentored 4 junior researchers in Python, feature engineering, and model evaluation best practices.

Financial Analyst | Creative Engineering Solutions, Peshawar *April 2023 – June 2025*

- Automated monthly financial reporting using Python and Power BI, cutting report generation time from 3 days to 4 hours.
- Engineered a cost anomaly detection model that identified £120,000 in recoverable operational inefficiencies within the first quarter of deployment.
- Developed and maintained 6 real-time dashboards used by senior management for strategic decision-making across 3 business units.
- Streamlined SQL data pipelines, shrinking data preparation time by 50% and improving reliability across reporting systems.
- Partnered with cross-functional teams to translate data insights into actionable business strategies.

PROJECTS

Deep Learning Tactical Asset Allocation *Python, PyTorch, Pandas, NumPy*

- Architected a Temporal Convolutional Network (TCN) with Attention mechanism for dynamic portfolio allocation across 30 DJIA constituent stocks.
- Outperformed all 5 traditional baselines (equal-weight, mean-variance, momentum, buy-and-hold, risk parity) on out-of-sample data from 2018 to 2024.
- Achieved a Sharpe ratio of 1.169 and 33.7% annualised return vs. 1.133 and 24.0% for the best baseline (mean-variance).
- Applied institutional-grade methodology: walk-forward validation with embargo periods, rolling normalisation with no look-ahead bias, and transaction costs embedded in the training loss.
- Final year project, BSc Data Science, University of the West of England, Bristol.

Medical Image Classifier

Python, PyTorch, FastAPI, Docker

- Trained a convolutional neural network to classify chest X-rays into 3 diagnostic categories (normal, pneumonia, nodule), achieving 91% accuracy on a held-out test set.
- Leveraged transfer learning with a pre-trained ResNet backbone and data augmentation, cutting training time by 60% vs. training from scratch.
- Deployed as a REST API via FastAPI and Docker, processing 200+ images per minute at sub-200ms inference latency.

Student Performance Prediction System

Python, Scikit-learn, Tableau

- Trained a Random Forest and Gradient Boosting model to identify at-risk students using academic and behavioural data from 8,000+ records across 3 academic years.
- Achieved 87% recall through targeted feature engineering (attendance patterns, submission timing, grade trajectory), enabling early intervention for a 22% improvement in support outcomes.
- Delivered findings via an interactive Tableau dashboard used weekly by academic advisors to prioritise student outreach.

Financial Anomaly Detection Dashboard

Python, SQL, Power BI

- Constructed an end-to-end anomaly detection pipeline using isolation forest and z-score methods to flag unusual spending patterns across company accounts.
- Integrated real-time alerting triggered when anomaly scores exceeded configurable thresholds, enabling same-day investigation of flagged transactions.
- Consolidated data from 4 disparate source systems into a single SQL reporting layer, replacing a manual weekly review process that took 6+ hours.

EDUCATION

BSc Data Science and AI (Top-Up) *(in progress)*

September 2025 – Present

University of the West of England, Bristol

- Advanced studies in deep learning, NLP, data analytics, and business intelligence with practical model deployment projects.

Bachelor of Data Science *(with Distinction)*

October 2022 – June 2025

Institute of Management Sciences, Peshawar

- Graduated in the top 5% of cohort.
- Dissertation at University of the West of England, Bristol: Deep Learning Tactical Asset Allocation, outperforming traditional portfolio strategies using TCN + Attention on DJIA stocks (2018 to 2024).

CERTIFICATIONS

- Deep Learning Specialization, DeepLearning.AI (Andrew Ng), 2021
- IBM Machine Learning Professional Certificate, Coursera, 2022
- Google Data Analytics Professional Certificate, Coursera, 2023
- Microsoft Certified: Azure Data Scientist Associate, 2024
- Bloomberg Market Concepts (BMC), Bloomberg, 2025
- Google Project Management Certificate, Coursera, 2025
- CFA Level 1 (in progress), CFA Institute, 2026

ADDITIONAL INFORMATION

Languages: English (fluent), Urdu/Hindi (fluent), Pashto (fluent)

Leadership & Community:

- Founded and led two student societies at Islamia College Peshawar: one focused on literary awareness and cultural engagement, the other on structured public discourse around contemporary political issues, organising regular events and panel discussions.
- Served as Class Representative throughout the majority of both school and university, acting as the primary liaison between students and faculty on academic and welfare matters.
- Volunteered in fundraising drives for Shaukat Khanum Memorial Cancer Hospital, Peshawar
- Regular blood donor

GitHub: github.com/zulfayqar-khan